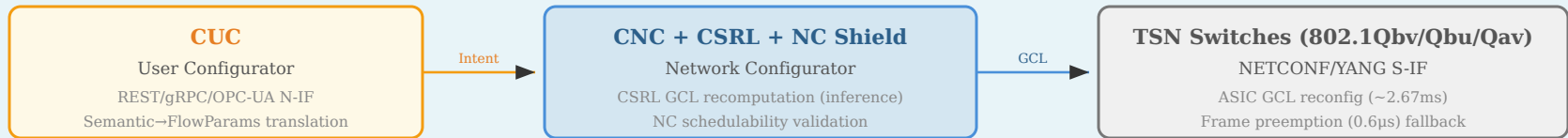


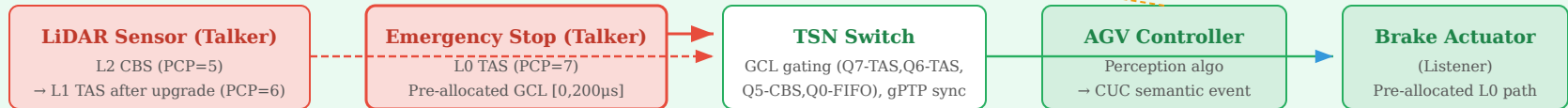
Semantic Signaling Architecture: CUC → CNC → GCL Reconfiguration

Control Plane (OAM VLAN, PCP=0, ms-scale)



(Control plane: non-real-time signaling, physically/logically isolated from data plane TAS/CBS queues)

Data Plane (TSN deterministic forwarding, μs-scale)



LiDAR frames (L2 CBS → L1 TAS after upgrade)

Processed data → brake command (L0, immediate)

CUC ← AGV: "LiDAR stream L2→L1" (ms-scale, non-real-time)